Neural Message-Passing on Attention Graphs for Hallucination Detection

Welcome, everyone. Today I'm presenting an approach to a critical problem in LLMs: hallucinations. While recent research has tried to detect these errors by probing internal LLM states—like looking at specific activation layers or attention scores—these methods treat the signals in isolation. This paper asks: what if we look at the holistic structure of how tokens interact during generation?

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The Core Innovation – CHARM

The core innovation here is framing hallucination detection strictly as a graph learning task. Instead of flattening the LLM's internal states into a sequence, CHARM constructs an attributed, directed graph. Here, every token generated is a node, and the attention mechanism dictates the edges connecting them. This allows the model to map the actual computational flow.

Method: CHARM

Catching Hallucinated Responses via Message-passing.

The Shift

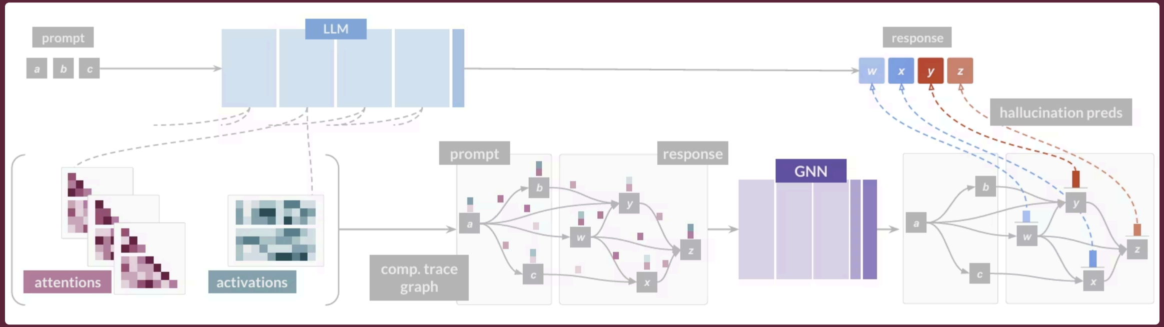
Unifying LLM traces by representing them as attributed directed graphs.

Graph Structure

Tokens are nodes, and the attentional flows between them form the edges.

Building the Computational Trace Graph

Let's look at the feature engineering. Nodes are enriched with their residual stream activations and self-attention scores. The edges carry the heavy lifting: vectors containing the attention weights across all layers and heads. Because attention matrices are dense and noisy, the authors apply a sparsification threshold to drop weak connections, keeping the graph computationally tractable without losing the primary signal.



Node Features

Token-wise signals: self-attention scores and residual stream activations.

Edge Features

Pairwise interactions: attention scores between distinct tokens across all layers and heads.

Sparsification

Edges with attention scores below a threshold are dropped for efficiency.

Why Graph Neural Networks (GNNs)?

If we represent tokens and attention as a graph, GNNs are the most principled tool to apply. This solves the granularity problem: if we want to know if a specific word is hallucinated, we perform node classification. If we want to evaluate the whole sentence, we do graph classification. Furthermore, the authors mathematically prove that a single message-passing layer can approximate prior state-of-the-art heuristic equations.

Flexibility

Accommodates response-level (graph classification) and token-level (node classification).

Integration

Naturally combines heterogeneous signals (attentions & activations).

Expressiveness

Provably subsumes prior hand-crafted attention heuristics.

Methodology – The CHARM Architecture

The CHARM pipeline is standard but highly effective. It starts with the message-passing function, which allows token embeddings to be updated based on their neighbors' features, weighted by the attention edges. For token-level detection, we push these straight to a prediction head. For response-level, we apply a pooling function—like a sum or average—before the final classification.

01

Message-passing

Stacks learnable layers to compute updated token representations from the graph.

02

Pooling

Aggregates representations into a single graph vector for response-level tasks.

03

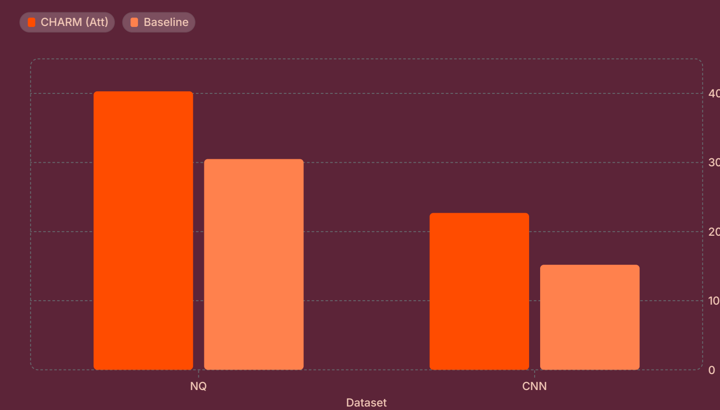
Prediction

Applies dense layers to output the final hallucination scores.

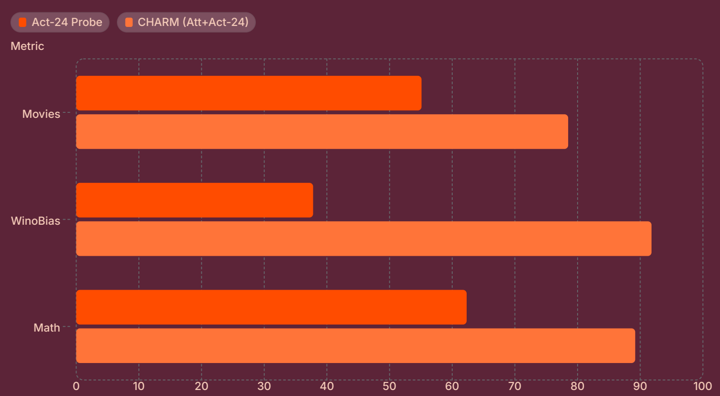
Contextual Hallucination Detection (Token-Level)

When evaluating token-level contextual hallucinations—where the answer is in the prompt but the model ignores it—CHARM consistently beats baselines. Interestingly, the model performed best using **only** attention features. Adding residual activations actually hurt performance here. The hypothesis is that attention flow alone contains the necessary signal for grounding, and adding activations just caused the model to overfit on spurious correlations.

- **Datasets:** NQ (Question Answering) and CNN (Summarization) with LLaMa-2-7B-chat.
- **Results (AUPR):** CHARM using only attention features achieved 40.3% on NQ and 22.7% on CNN, outperforming standard probes.



Response-Level Detection (Factual & Reasoning)



The narrative flips when we look at response-level tasks requiring internal knowledge or logic, like Math or factual recall. Here, combining attention with residual stream activations creates a powerful synergy. On the WinoBias dataset, for example, the combination pushed the AUPR from roughly 38% using just activations to nearly 92%. This proves the value of a flexible framework that can ingest heterogeneous node and edge data.

- **Datasets:** Movies (factual recall), WinoBias (bias), Math (arithmetic reasoning) with Mistral-7B-instruct.
- **Results:** CHARM combining attention and layer 24 activations achieved massive gains.

Efficiency and Robustness

A major concern with graph methods is computational overhead. However, CHARM is highly robust to sparsification. By thresholding attention weights at 0.05, we drop over 90% of the edges but maintain top-tier performance. Because it only requires a single forward pass of the LLM to extract these traces, CHARM executes in milliseconds—making it far superior to methods that require sampling dozens of alternative text responses.

>90%

Sparsity Tolerance

Performance remains stable even at threshold 0.05, removing most edges.

ms

Inference Latency

Extremely fast, enabling real-time detection.

~5.9s

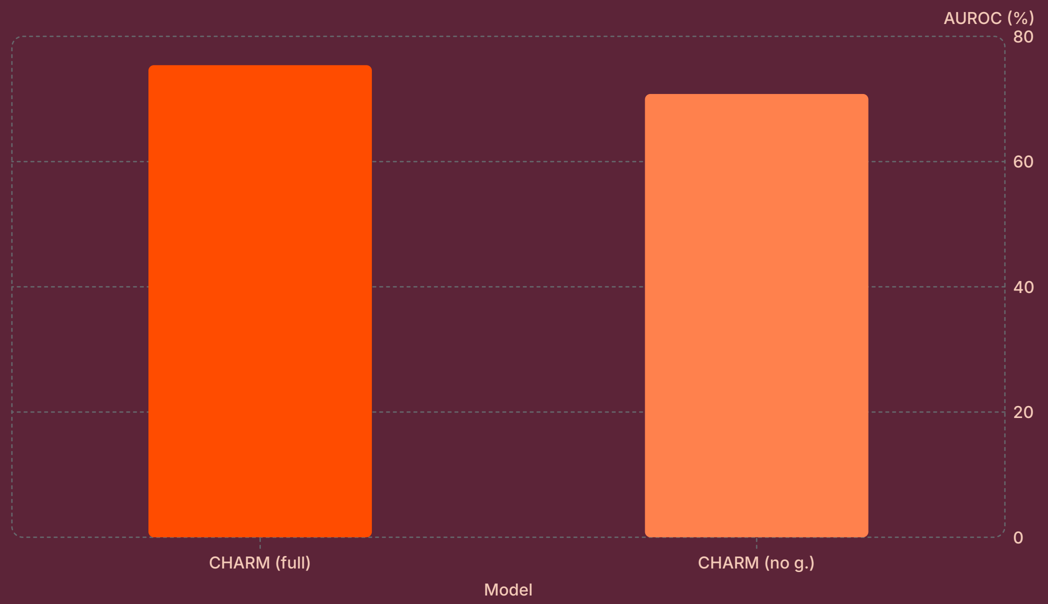
Alternative Methods

Orders of magnitude slower than multi-prompting methods.

Ablation Study – The Importance of the Graph

To prove that message passing is actually doing the work, the authors ran an ablation study where they completely severed the graph connectivity. The model was reduced to just processing node features with dense layers. Across the board, performance dropped significantly. This confirms that the relational structure—how tokens attend to one another—is strictly necessary for accurate detection.

- **Test Setup:** Comparing full CHARM against "CHARM (no g.)", replacing message-passing with standard dense layers over isolated node features.
- **Results:** Removing the graph structure consistently drops performance.
- **CNN AUROC:** Drops from 75.4% to 70.8% without the graph.



Thank You

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